Shreya Nakum

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Education

University of California, Irvine Sep 2023 - Jun 2027
Bachelor of Science in Computer Science with Honors (GPA: 3.82 / 4.00) Irvine, California

- **Honors/Awards:** 1st Place Best Collaborative Research at UCI UROP 2025, Chancellor's Award for Excellence in Undergraduate Research, HackMIT Modal Honorable Mention, 6x Dean's Honors List.
- **Organizations:** AI@UCI, Campus Honors Collegium, Information & Computer Science Honors Program.

Experience

Jet Propulsion Lab (JPL) Oct 2026 – May 2027
Robotics & Vision Intern (358T) Pasadena, CA

- Building an autonomous driving algorithm for the Scarecrow rover.

NASA: Johnson Space Center Jan 2026 – Aug 2026
SWE & GN&C Intern (EG4) Houston, Texas

- Confirmed the stability of non-propulsive altitude maneuvers on LEO stations by developing reaction wheel actuator models in MATLAB/Simulink.
- Added reaction wheel utilities into internal high-fidelity altitude dynamics simulations in Fortran 77.
- Improved lander module altimeters by developing a laser-detection computer vision algorithm utilizing ROS and OpenCV.
- Presenting computer vision algorithm work at the 49th American Astronautical Society GN&C conference in Breckenridge.

UC Irvine: Donald Bren School of Information & Computer Science Dec 2025 – Present
Researcher (Advisor: Dr. Krone-Martins) Irvine, CA

- Identified gaps in training for LLMs by graphically evaluating physics & math datasets using PacMAP.
- Determined model proclivity and manifold quality by calculating Davies-Bouldin indices, mean silhouette scores, and intra-cosine similarity using scikit-learn.

National Science Foundation: University of Oulu - M3S Lab June 2025 – Aug 2025
Researcher (Advisor: Dr. Matteo Esposito) Oulu, North Ostrobothnia, Finland

- Built a RAG pipeline to measure code clone similarity in software repositories with vllm, ollama, and HuggingFace.
- Benchmarked 3 LLMs across zero-shot, few-shot, and chain-of-thought prompts with 3k+ thresholding values.
- Automated statistical evaluation of model metrics; publishing a research paper to ASEJ.

UC Irvine: Rose Labs (School of Biological Sciences) Sep 2024 – Present
Researcher (Advisor: Dr. Michael Rose) Irvine, CA

- Built an automated Python machine learning pipeline to train and evaluate 220 supervised classification models, achieving 80% accuracy in evolving DNA sequence identification and reducing manual genomic analysis time by 32%.
- Increased assay tracking reliability by 10 times by engineering a 2D pose estimation and identity algorithm
- Deployed a data annotation (Flask+JS) application that accelerated data collection by 10x.
- Designed a CNN object detection pipeline & trained a model (95% accuracy) to count *Drosophila melanogaster* eggs.
- Awarded 2 research prizes for novel small-object detection methods.

California Institute of Technology: Theoretical AstroPhysics Including Relativity and Cosmology (TAPIR) Oct 2021 – Mar 2023
Researcher (Advisor: Dr. Iryna Butsky) Pasadena, CA

- Processed Hubble COS-Halos datasets with Python (astropy, numpy, pandas).
- Derived galaxy halo mass and velocity profiles; visualized results with matplotlib.
- Published findings in *MNRAS* and presented it at the American Astronomical Society conference.

Publications

Nakum, S., Kemman, J., Esposito, M., Cerny, T., Lenarduzzi, V., & Taibi, D. (2026). An LLM-based Technique for Detection of Code Clones Across Open Source Microservice Repository Forks. *Automated Software Engineering Journal*, **In Review.** https://doi.org/10.13140/RG.2.2.34393.28004

Butsky, I., Nakum, S., Ponnada, S., Hummels, C., Ji, S., & Hopkins, P. (2022). Constraining Cosmic-ray Transport with Observations of the Circumgalactic Medium. *Monthly Notices of the Royal Astronomical Society*, **521**(2), 2477–2483. https://doi.org/10.1093/mnras/stad671

Projects

HackMIT: ActsAI | *Honorable Mention, Modal.AI* GitHub

- Built a mental health platform with a Llama-based chatbot to connect users to therapists; integrated SMS alerts for crises.

Atoma Media: Financial News Sentiment Analysis HuggingFace

- Built and automated Twitter scraper with PostgreSQL & AWS Lambda to collect financial news headlines and comments.
- Fine-tuned a sentiment-analysis model to predict financial news sentiment, achieving 80% accuracy and 6k+ downloads.

Technical Skills

Languages: Python, C++, C, R, Javascript/Typescript, SQL, MATLAB, Fortran 77.
Frameworks & Libraries: PyTorch, TensorFlow/Keras, Flask, FastAPI, scikit-learn, transformers, Robot Operating System (ROS).
Tools & Platforms: AWS (Lambda, Boto3), Docker, PostgreSQL, Milvus, HuggingFace, Git, WordPress, Simulink, Linux.
Expertise: Computer vision, RAG, Data Analysis, Physical Simulations, Academic Writing.
Other: Web Development, API Integration, Web Scraping, MLOps, Supercomputing.